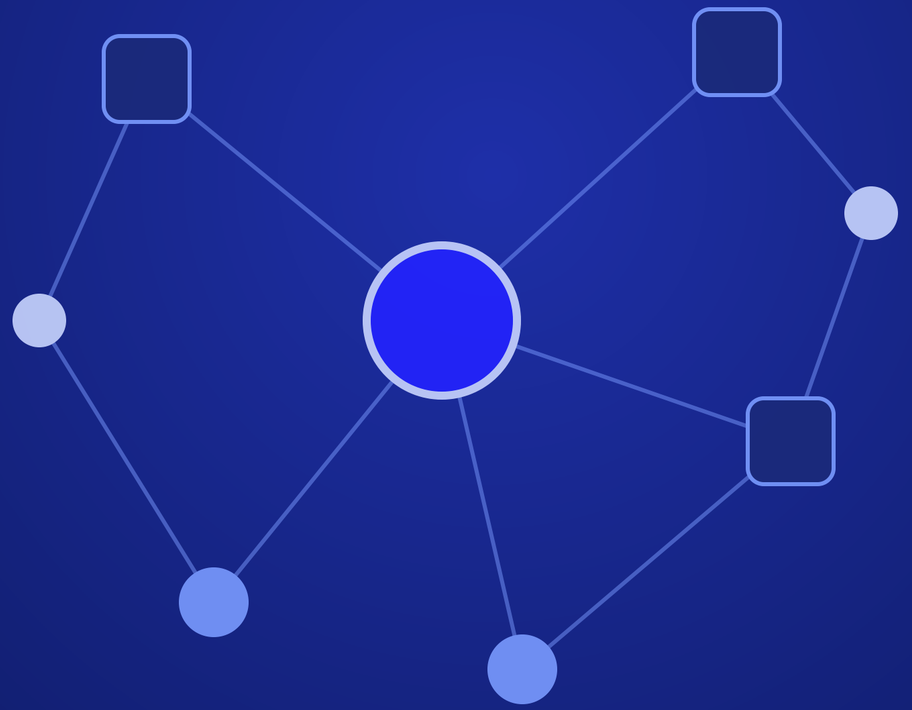
kaiku

What your experts know, ready for every agent, at the right moment.

The time-aware context layer for network operations.



Elisa × SinceAI Hackathon 2026

Problem

noc-transport Thread · TRN-042

VV **Veikko Vikkelä 09:14**
Changed the SFP on transmission node TRN-042. Link is up and alarms cleared, but traffic is still not coming back.

VV **Veikko Vikkelä 09:18**
So another SFP?

EE **Erkki Expert 09:22**
That’s it. Disable the interface, wait a little, then enable it again. Don’t reboot the whole unit unless that fails.

VV **Veikko Vikkelä 09:26**
Is this documented somewhere?

EE **Erkki Expert 09:27**
Not that I know of 😄 We’ve just learned it over the years.

An AI agent on the same fault

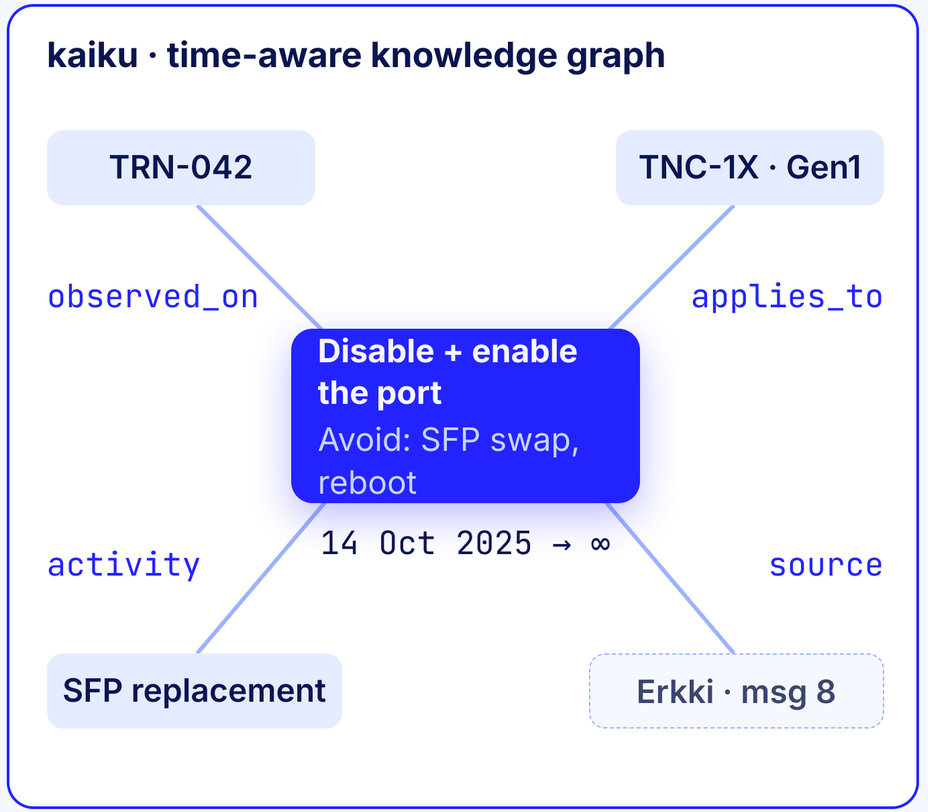
- **Detect**
- **Diagnose**
- **Decide**
  Context missing
- **Fix**
  Another SFP, or a reboot

Solution

Where experts talk

- Slack
- Teams
- Email
- PDFs, wiki, KB

capture



serve

MCP server /mcp

```
resolve_entities
get_entity
get_context
search_context
check_conditions
get_site_access
get_events
get_history

report_outcome
open_knowledge_gap
propose_assertion
flag_assertion
```

Any agent: ours, Claude, LangGraph

Tech stack



Python



FastAPI



Pydantic



LangGraph



MCP



Neo4j



DuckDB



Phoenix



React



TypeScript



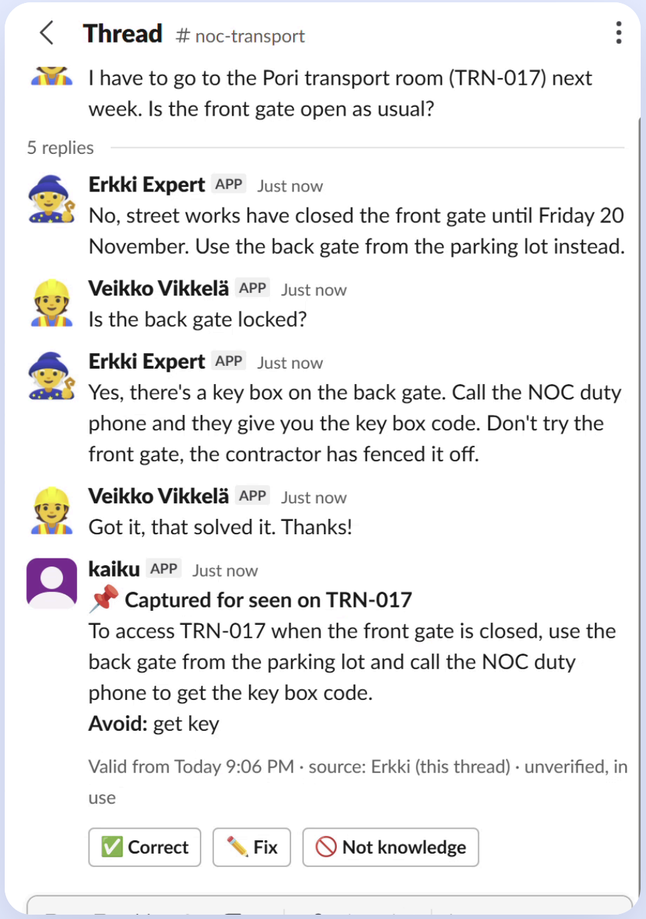
Vite

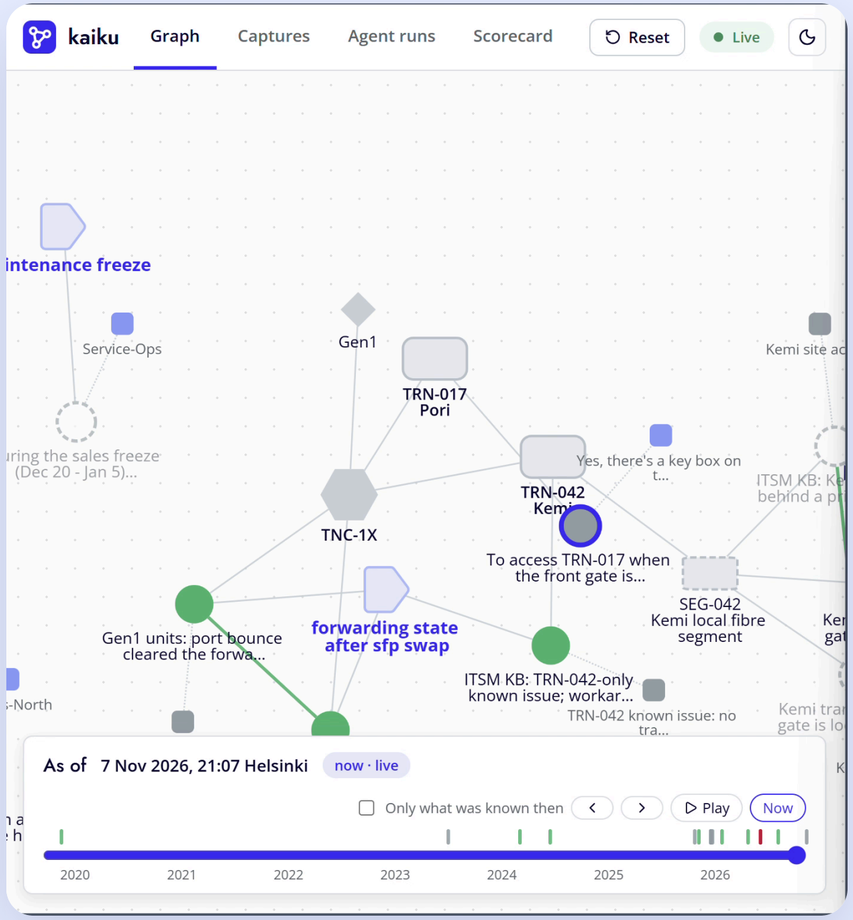


Docker

Models via Featherless: Kimi-K2 (extraction, agent) · Qwen3-4B (triage) · Qwen3-Embedding-8B (search)

Capture pipeline



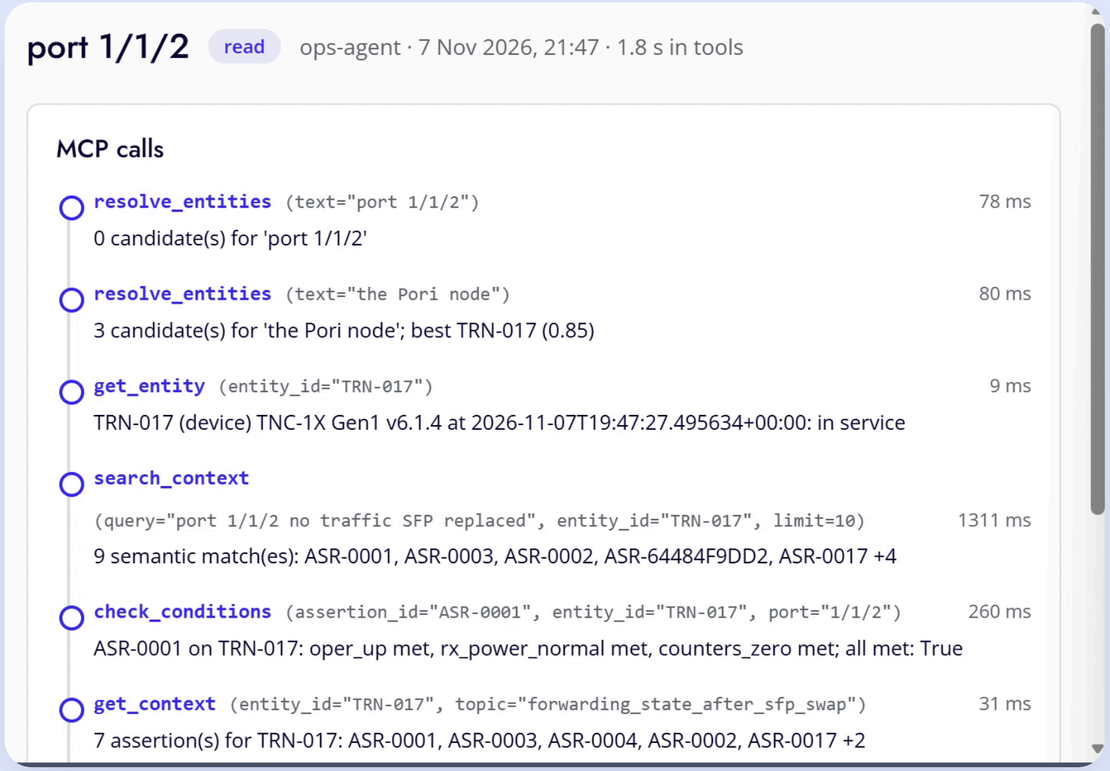


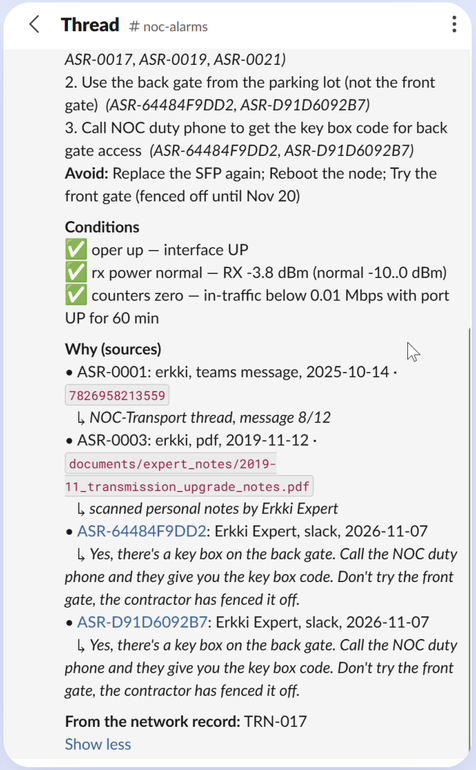
1. Veikko asks about the Pori site; Erkki answers in the thread.
2. kaiku reads the thread on its own and runs the pipeline.
3. It posts one card: captured for TRN-017.
4. The graph adds the node, valid until 20 Nov.
5. “Correct” is optional; one click confirms it.

Live recording, 7 Nov 2026 · Slack (#noc-transport) and the kaiku console · synthetic network data

Retrieval pipeline

@kaiku *“Port 1/1/2 on the Pori node has had no traffic since its SFP was replaced. What should I do, and how do I get in?”*





Live recording, 7 Nov 2026 · kaiku console Agent runs (MCP trace) and the ops agent’s Slack reply · synthetic network data

Features

Time-aware knowledge graph

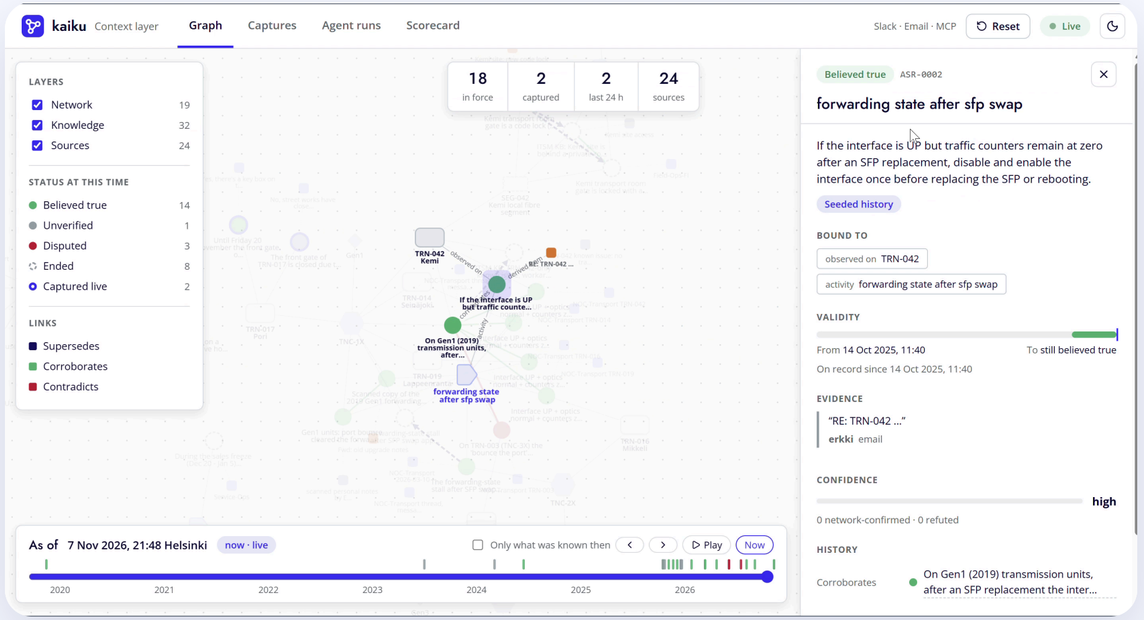
Every fact carries valid_from, valid_to and recorded_at. Query any instant; nothing is overwritten.

Automated extraction and retrieval

Chat and email in, structured facts out, every field quoted from its source. Retrieval filters by time, then ranks.

Agent-ready MCP

12 tools any agent can call. Agents add evidence; only people and the network confirm.



"As of" shows what was in force at any date; "Only what was known then" also hides what was recorded later.

How capture works

Triage keep or ignore → **Extract** rule + quotes → **Link** entities + dates → **Validate** quality checks → **Reconcile** merge with known → **Persist** Neo4j + Slack card

Extract · every field is a quote from the thread

Source

action avoid condition entity

Veikko Vikkelä 21:25 · open
I have to go to the Pori transport room (TRN-017) next week. Is the front gate open as usual?

Erkki Expert 21:25 · open
No, street works have closed the front gate until Friday 20 November. Use the back gate from the parking lot instead.

Veikko Vikkelä 21:25 · open
Is the back gate locked?

Erkki Expert 21:25 · open
Yes, there's a key box on the back gate. Call the NOC duty phone and they give you the key box code. Don't try the front gate, the contractor has fenced it off.

Veikko Vikkelä 21:25 · open
Got it, that solved it. Thanks!

Link · bound to the graph, with a validity window

`ASR-64484F9DD2 · site_access`

Until Friday 20 November the front gate of the Pori transport room (TRN-017) is closed for street works and fenced off by the contractor. Use the back gate from the parking lot: it has a key box, and the NOC duty phone gives out the key box code.

bound to	`TRN-017 · observed_on`
action	`use_back_gate`
avoid	`front_gate`
valid	`7 Nov 21:25 → 20 Nov 00:00`

How retrieval works

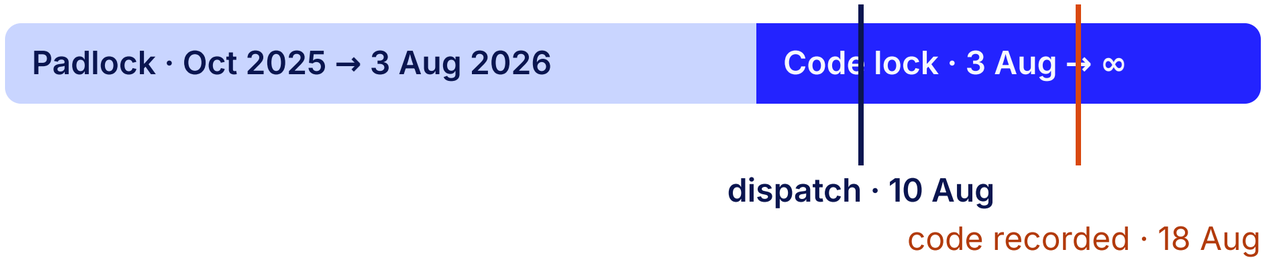
1 Resolve
"the Pori node" → TRN-017 (0.85): TNC-1X, Gen1, fw v6.1.4

2 Filter at t
Valid at t, device existed at t, model and firmware match

3 Rank
Meaning × trust, over the valid set only

4 Check & cite
Conditions vs live telemetry; outcome fed back

Why the agent sent an engineer to Kemi without the code



as_of 10 Aug → code lock: what was true.
+ known_at 10 Aug → padlock: what we knew then.

```
score =
max(0, cos(q, a))
× scope × status
× confidence
```

Scope: node › model › generation › site › segment.

No age decay. A replaced fact always shows its successor.

Impact

Without context

- Agent: “Optics fine, no traffic: replace the SFP.”
- Second SFP swapped. Still no traffic.
- Engineer reaches the site: front gate locked.
- Workflow stops. A human has to step in.

With kaiku

- Agent: “Matches Erkki’s rule: disable and re-enable port 1/1/2.”
- Traffic back. No hardware swapped.
- Back gate, key box code from the NOC duty phone.
- Fixed, end to end.

Faster fault resolution

The right next step before hardware is replaced or the unit is rebooted.

Better autonomous ops

Workflows continue end to end instead of waiting for a human.

Knowledge retention

Expert know-how captured from the channels they already use.